

Factors distinguishing treatment-seeking individuals and individuals who self-identified as coping well following the Canterbury Earthquake Sequence

Cate F. Woods¹,
Virginia V. W. McIntosh¹,
Christopher M. Frampton²,
Frances A. Carter⁴,
Janet D. Carter¹,
Helen C. Colhoun³,
Jennifer Jordan^{2,3},
Rebekah A. Smith³,
Caroline Bell²

¹ School of Psychology, Speech and Hearing, University of Canterbury, Christchurch, New Zealand

² Department of Psychological Medicine, University of Otago, Christchurch, New Zealand

³ Te Whatu Ora – Waitaha (Health New Zealand Canterbury), Christchurch, New Zealand

⁴ Formerly Te Whatu Ora – Waitaha Canterbury (Health New Zealand), Christchurch, New Zealand (not currently working)

© The Author(s) 2025. (Copyright notice)

Author correspondence:

Cate Woods

E-mail: cate.woods@pg.canterbury.ac.nz.

Abstract

Following potentially traumatic events, most individuals do not develop mental health issues and some even experience high levels of psychological functioning. Others, however, develop mental health conditions such as posttraumatic stress disorder (PTSD), depression, or anxiety disorders, and require specialist mental health treatment. Few studies have directly compared individuals at these opposite ends of the trauma response spectrum following the same event. The current study identified factors distinguishing individuals seeking treatment for severe earthquake-related distress (n = 184) and those who did not develop earthquake-related distress and self-identified as coping well (n = 101) following the 2010/11 Canterbury earthquake sequence. Participants completed measures of sociodemographic

factors, mental health history, earthquake exposure, life events, and current psychological functioning. In univariate analyses, the groups reported comparable levels of objective earthquake exposure, but treatment-seeking participants reported significantly more pre-earthquake mental disorders, peritraumatic distress (i.e., emotional and physiological distress experienced during the earthquake), exposure-related distress (i.e., distress associated with earthquake-related traumatic events), higher number of life events (e.g., marriage, death of a family member) experienced in the past 5 years, and greater perceived difficulty associated with life events. Treatment-seeking participants were also significantly younger, more likely to be female, less likely to have a tertiary educational qualification, and less likely to be in a relationship. In multivariate analyses, peritraumatic distress and education level were the only factors that uniquely distinguished the groups. Findings emphasise the importance of peritraumatic experiences and education level in the development of distinct post-trauma outcomes among individuals exposed to similar degrees of disaster exposure. Subjective peritraumatic responses are likely to be useful to consider in screening efforts following disasters.

Keywords: Posttraumatic stress disorder, post-trauma outcomes, disaster, earthquake

Over 70% of people are exposed to potentially traumatic events during their lifetime (Benjet et al., 2015). Trauma exposure increases risk for mental disorders, including posttraumatic stress disorder (PTSD), mood, anxiety, and substance use disorders (Auxéméry, 2018). There has been a focus on PTSD following potentially traumatic events, which is characterised by intrusion, avoidance, and hyperarousal symptoms, alongside disturbed cognition, mood, and functioning (Asmundson et al., 2000). Despite the high prevalence of trauma exposure, relatively few individuals develop PTSD (lifetime prevalence is 8.3%; Kilpatrick et al., 2013) and most people experience only mild trauma responses, with few or no PTSD symptoms (Bonanno, 2004).

A range of pre-, peri-, and post-trauma factors have been found to influence risk for the development of PTSD. Pre-trauma factors are pre-existing characteristics unrelated to the traumatic event, including sociodemographic factors and factors relating to pre-trauma mental health and trauma exposure. Female gender is a robust risk factor for the development of PTSD (Tortella-Feliu et al., 2019). Women are more likely than men to experience events conferring greater risk for PTSD (e.g., sexual assault) and are more likely to experience other risk factors, such as heightened peritraumatic distress (i.e., emotional and physiological distress experienced during the traumatic event; Christiansen & Hansen, 2015). Other sociodemographic factors, including younger age and lower education, were identified as risk factors in earlier meta-analyses (Brewin et al., 2000), but these findings have not been replicated in larger, more recent syntheses (Tortella-Feliu et al., 2019). Higher rates of PTSD have also been observed among ethnic minority groups (Asnaani & Hall-Clark, 2017). The presence of pre-trauma mental disorder, particularly prior PTSD, is a well-established risk factor for PTSD after subsequent exposure (Brewin et al., 2000; Fernandez et al., 2016). Prior trauma exposure has also been found to increase risk for PTSD (Ozer et al., 2003).

Peritraumatic factors relate to an individual's experiences during and immediately after the traumatic event, including the severity of objective trauma exposure and subjective responses to the event. Greater degree of objective exposure is a risk factor for PTSD (Brewin et al., 2000). In the context of a disaster, for example, people who experienced bereavement, sustained injuries, or lost possessions are considered to have a higher degree of objective exposure (Elal & Slade, 2005). The importance of subjective experiences has been noted in addition to objective experiences. People who experience more intense subjective emotional and physiological distress during a traumatic event (i.e., greater peritraumatic distress, panic, and dissociation) are more likely to develop PTSD (Vance et al., 2018). It is hypothesised this may be due to the way these responses disrupt the encoding of the event into memory and contribute to negative post-trauma cognitions (Ehlers & Clark, 2000).

Post-trauma factors are those that occur after the traumatic event, including degree of post-trauma social support, life stress, and resource loss. Reduced social support has been identified as a risk factor for PTSD in earlier meta-analyses (Ozer et al., 2003); however, this finding has not been replicated in more recent syntheses (Tortella-Feliu et al., 2019). Increased life stress following

trauma exposure is a risk factor for PTSD (Brewin et al., 2000), perhaps because stress depletes the psychosocial resources people utilise to cope with traumatic events. Relatedly, conservation of resources theory (Hobfoll et al., 2006) posits that people who lose a greater degree of resources following a traumatic event (e.g., personal possessions, psychological coping resources, valued social roles) are more likely to develop PTSD.

Disasters triggered by natural processes are collectively experienced and often expose entire communities to a series of potentially traumatic events. Individuals exposed to the same disaster may have similar objective experiences but can experience markedly different outcomes. Some individuals do not develop disaster-related mental health issues and, additionally, report high levels of psychological functioning despite substantial disaster exposure (Bonanno et al., 2006). Others, however, will require specialist mental health treatment for severe and ongoing disaster-related distress (North & Pfefferbaum, 2013). While multiple studies have examined factors associated with the development of PTSD symptoms among people exposed to the same disaster, few studies have directly compared individuals at opposite ends of the trauma response spectrum: those seeking treatment for disaster-related distress and those who exhibit high levels of functioning and the absence of mental health problems, despite substantial levels of exposure severity. Factors distinguishing these groups could be used to identify individuals most at risk of requiring specialist mental health services, which has potential to inform outreach and service planning efforts following disasters.

In 2010 and 2011, a series of major earthquakes occurred in the Canterbury region in Aotearoa New Zealand, affecting a large proportion of the community and resulting in 185 deaths, thousands of injuries, and widespread damage (Ardagh et al., 2012). Many Canterbury residents coped well, but others sought treatment for severe, ongoing, earthquake-related distress. The current study examined factors distinguishing people who sought treatment compared to a control group who did not develop mental health issues and coped well following the Canterbury earthquakes, including factors relating to sociodemographics, pre-earthquake mental health, degree of trauma exposure, peritraumatic distress, and life events.

Method

Study Design

The study utilised an observational cross-sectional design, in which earthquake-exposed participants with distinct post-earthquake outcomes were retrospectively assessed at one time point following the earthquakes.

Participants

Earthquake-Exposed Treatment-Seeking Group

This group comprised 184 Canterbury residents referred to a specialist mental health service that was initiated for the treatment of severe, ongoing earthquake-related distress (i.e., PTSD or adjustment disorder with earthquake-related anxiety). The service was the main provider of publicly-funded mental health treatment for those with severe levels of distress related to the earthquakes. Participants were referred by general medical practitioners, other mental health services, or psychiatrists and psychologists working in private practice. People were invited to take part in the study during initial face-to-face assessments (see Procedure section), but participation was not required to receive treatment. Inclusion criteria were age 18 years and above and presence of earthquake-related DSM-IV PTSD or adjustment disorder with earthquake-related anxiety. Those for whom the primary earthquake-related disorder was not PTSD or adjustment disorder (e.g., depression) were not treated in the service and were referred to existing tertiary mental health services. Other exclusion criteria were current mania, psychotic depression, severe alcohol or drug disorders/problems, and active suicidality. Treatment-seeking participants were recruited over a period of 3 years and 8 months, between August 2011 and April 2015, approximately 1 to 4.5 years after the onset of the earthquake sequence. The flow of treatment-seeking participants through the recruitment process is displayed in Appendix A. Of the 253 people who were referred to the service and attended initial clinical assessments, 184 (72.7%) were suitable for the service and consented to participate in the study.

Earthquake-Exposed Control Group

This group comprised 101 Canterbury residents who did not develop earthquake-related mental health issues and who self-identified (at the time of recruitment) as having coped well following the earthquakes, despite moderate to high exposure to earthquake-related events such as sustaining physical injury, seeing buildings falling down, seeing injured people and dead bodies, or damage to

property and possessions (henceforth referred to as the “control group” or “control participants”). Control participants, recruited from October 2012 to November 2013, responded to publicity in newspapers, flyers posted in public places, community websites, and word of mouth. Potential participants were invited to respond by contacting the research team directly, either by telephone or email. Control participants were excluded if they had developed an earthquake-related mental disorder or received treatment for earthquake-related distress. Participants were not excluded if they had current mental disorders that were unrelated to and pre-dated the earthquake sequence. The research team met regularly to discuss and determine the eligibility of control participants, after both the initial screening and interview assessments. Those who did not experience substantial earthquake exposure or who were assessed as not coping well were excluded. The flow of control participants through the recruitment process is displayed in Appendix B.

Procedure

Both control and treatment-seeking participants were screened by telephone for eligibility before attending face-to-face assessments, during which participants were provided with verbal information about the study alongside a written information sheet. Once participants had received information about the study, informed written consent was obtained prior to data collection. The face-to-face assessment consisted of the Mini International Neuropsychiatric Interview (MINI; Sheehan et al., 1998) to assess the presence of mental disorders. Participants also completed pen-and-paper demographic and self-report measures (described below). Participants were allowed to skip questions they did not want to answer. Treatment-seeking participants were assessed by experienced mental health clinicians (e.g., clinical psychologists, experienced mental health nurses); control participants were assessed by research assistants or post-graduate students trained to administer the MINI. Participants did not receive incentives for participation.

Measures

Sociodemographic Information. Participants provided sociodemographic information relating to gender, age, relationship status, education, and ethnicity.

Mini-International Neuropsychiatric Interview (MINI). The MINI is a structured diagnostic interview used to assess the presence of common mental disorders including PTSD, adjustment disorder, mood, anxiety,

substance use, and eating disorders (Sheehan et al., 1998). Current and pre-earthquake diagnoses were assessed. Pre-earthquake diagnoses were based on post-earthquake recall.

Posttraumatic Stress Disorder Checklist—Specific (PCL-S). The PCL-S is a 17-item scale assessing DSM-IV PTSD symptoms relating to a specific event (in this case, the Canterbury earthquakes; Blanchard et al., 1996). Five items assess intrusion symptoms ($\alpha = .95$; this value, and subsequent α values, are from the current sample), seven items assess avoidance symptoms ($\alpha = .91$), and five items assess hyperarousal symptoms ($\alpha = .93$). Participants rate items from 1 = *not at all* to 5 = *extremely*. A total severity score was calculated by summing all item scores ($\alpha = .97$; possible range = 17–85; this and all other ranges reported in this section reflect possible ranges for the scale). Higher scores reflect more severe PTSD symptoms.

Connor-Davidson Resilience Scale (CD-RISC). The CD-RISC (Connor & Davidson, 2003) is a 25-item scale assessing components of resilience such as adaptability to change, engaging the support of others, and self-efficacy. Participants rate items from 0 = *not true at all* to 4 = *true nearly all the time*. Item scores are summed to produce a total score ($\alpha = .95$; range = 0–100). Higher scores reflect greater resilience.

Depression Anxiety Stress Scales (DASS-21). The 21-item DASS-21 measures past-week depression, anxiety, and stress (Lovibond & Lovibond, 1995). Seven items pertain to each of the depression ($\alpha = .95$), anxiety ($\alpha = .93$), and stress ($\alpha = .95$) subscales. Items are rated from 0 = *did not apply to me at all* to 3 = *applied to me very much, or most of the time*. Subscale items are summed and multiplied by two to produce subscale scores (range = 0–42). Higher scores indicate greater depression, anxiety, and stress.

Social Adjustment Scale (SAS). The SAS is a 45-item scale measuring social functioning over the past 2 weeks (Weissman & Bothwell, 1976) across seven domains: work outside the home (six items; $\alpha = .92$), household tasks (six items; $\alpha = .83$), social and leisure activities (nine items; $\alpha = .83$), extended family (seven items; $\alpha = .64$), partner (ten items; $\alpha = .94$), parental (four items; $\alpha = .94$), and family unit (three items; $\alpha = .88$). Items are rated from 1 to 5. A total score reflecting social adjustment across all domains was produced by averaging all item scores ($\alpha = .91$; range = 1–5). Higher scores indicate poorer social adjustment.

Buss-Perry Aggression Questionnaire (BPAQ).

The current study used a shortened 12-item version (Bryant & Smith, 2001) of the BPAQ (Buss & Perry, 1992) to measure aggression. The measure contains four subscales, each containing three items: physical aggression ($\alpha = .70$), verbal aggression ($\alpha = .89$), anger ($\alpha = .86$), and hostility ($\alpha = .91$). Items are rated from 1 = *extremely uncharacteristic of me* to 5 = *extremely characteristic of me* and summed to produce a total score ($\alpha = .91$; range = 12–60). Higher scores indicate greater aggression.

Traumatic Exposure Severity Scale (TESS). Degree of earthquake exposure was assessed using a modified version of the TESS (Elal & Slade, 2005). The original TESS is a 24-item measure assessing resource loss, damage to home and goods, personal harm, concern for significant others, and exposure to the grotesque. Items were removed, modified, and added to reflect the experiences of Canterbury residents. The modified version deleted one item (“Did you experience the odour of dead bodies in the days following the earthquake?”) and split one item (“Were you buried under rubble/trapped for a period of time”) into two items (“Were you buried under rubble for a period of time after the earthquake?”, “Were you trapped in a building for a period of time after the earthquake?”). Six items were added relating to acute exposure (e.g., “Did you see buildings falling down as a result of the earthquake?”) and nine items were added relating to secondary stressors (e.g., “Have you lost your job since the earthquake?”), creating a 39-item measure. For each item, participants indicate whether they experienced the event, and if so, how distressing the event was (1 = *not at all* to 5 = *extremely*). An occurrence score was produced by summing the number of events experienced. A distress score was produced by summing distress scores. Higher occurrence and distress scores reflect greater earthquake exposure and exposure-related distress, respectively.

Peritraumatic Distress Inventory (PDI). The PDI is a 13-item scale measuring distress during and immediately after a traumatic event (Brunet et al., 2001). The measure includes items pertaining to both emotional distress (e.g., “I felt helpless”; “I felt afraid for my own safety”) and physiological distress (e.g., “I had physical reactions like sweating, shaking, and my heart pounding”; “I felt I might pass out”). Participants identified the earthquake they found most distressing as the index event. Items are rated from 0 = *not at all* to 4 = *extremely true*. The total score is the mean item score ($\alpha = .92$; range = 0–4). Higher scores reflect greater peritraumatic distress.

Life Events Scale (LES). An adaptation of the Crisis in Family Systems Revised Questionnaire (Shalowitz et al., 1998) was used to measure number of life events experienced in the past 5 years. The measure includes 64 items relating to 11 domains (e.g., financial, legal, career). Two items were added in the adapted version ("Did a family pet die?" and "Did a close family member divorce or separate?"). Total number of life events was calculated by summing each event experienced. The perceived difficulty of experienced events was rated from 1 = *not at all* to 4 = *a lot*. A total difficulty score was the mean of the difficulty ratings. For treatment-seeking participants recruited at the onset of the mental health service, the 5-year period mostly covered events occurring before the earthquakes. For those recruited towards the end of the service, the 5-year period mostly covered events occurring after the earthquakes.

Ethical Approval

The study was approved by the National Health and Disability Ethics Committee (URA/11/EXP/027, URA/12/03/011).

Data Analysis

Descriptive Analyses

All statistical analyses were conducted using SPSS v. 28.01. Demographic and MINI data were summarised using standard descriptive statistics including means and standard deviations for continuous measures and frequencies and percentages for categorical measures.

Group Differences in Current Psychological Functioning

Univariate logistic regression analyses were used to examine group differences on measures of current psychological functioning. These analyses were conducted to ensure that the treatment-seeking and coping well groups were indeed different in terms of post-earthquake psychological functioning (as intended), and to provide a descriptive overview of the groups' current psychological functioning. Posttraumatic stress symptoms (PCL), resilience (CD-RISC), depression, anxiety, and stress symptoms (DASS), social functioning (SAS), and aggression (BPAQ) were individually entered as independent variables, with group membership (treatment-seeking vs. control) as the dependent variable.

Main Analyses

A series of univariate logistic regression analyses was first used to identify factors distinguishing treatment-seeking and control participants. Number of pre-earthquake

mental disorders (MINI), objective earthquake exposure severity and exposure-related distress (TESS), peritraumatic distress (PDI), number of life events experienced in the past 5 years and difficulty associated with life events (LES), and sociodemographic factors (gender, age, education level, and relationship status) were each individually tested for association with post-trauma outcome (treatment-seeking vs. control). Univariate analyses were conducted prior to multivariate analyses to understand the association between individual factors and post-trauma outcome, and to identify candidate variables for multivariate analyses.

Factors identified as potentially associated with post-trauma outcome (using a significance threshold of $p < .10$) were considered candidate predictors. A higher significance threshold (i.e., $p < .10$ rather than the conventional $p < .05$) was used when identifying candidate predictors, as the primary aim of univariate analyses was to identify potential relationships, rather than to draw inferences about factors independently distinguishing the treatment-seeking and control groups. Multivariate logistic regression models were then constructed to estimate associations between candidate predictors and post-trauma outcome (treatment-seeking vs. control). To identify unique predictors, candidate predictors were entered into two separate regression models using forwards and backwards stepwise approaches. Predictors significantly associated with post-trauma outcome ($p < .05$) in both models were then entered simultaneously into a final logistic regression model. To account for differences in participants' timing of assessment, the final model was re-run adjusting for time elapsed between the onset of the earthquakes (04 September 2010) and date of assessment (less than 2.5 years vs. more than 2.5 years). To assess performance of the final model, predicted probabilities were generated from this model and used to calculate the area under the receiver operating curve (AUROC) predicting post-trauma outcome.

Missing Data

In the treatment-seeking group, missing data rates for variables ranged from 0% to 38.6%. These participants came from a clinical service operating in a post-earthquake environment and it was not always possible to collect full data for each participant. In the control group, missing data rates for variables ranged from 0% to 15.8%. In univariate logistic regression analyses, all cases with valid values on the predictor of interest were included. In multivariate analyses, a listwise deletion approach was implemented whereby any cases with missing values were excluded from analysis. Sample sizes for the individual measures are shown in tables.

Results

Descriptive Analyses

In the treatment-seeking group ($n = 184$), 153 participants (83.2%) were female, 30 (16.3%) were male, and one (0.5%) identified as a transgender woman. The mean age was 43.7 years ($SD = 13.27$). Ninety-six participants were in a relationship (52.5%; one did not have available data for relationship status) and 55 had a tertiary educational qualification (36.2%; 32 did not have available data for education level). Most treatment-seeking participants reported New Zealand European ethnicity ($n = 161$, 87.5%). In the control group ($n = 101$), 67 participants (66.3%) were female, and 34 (33.7%) were male. The mean age was 50.1 years ($SD = 11.02$). Seventy participants were in a relationship (69.3%) and 77 had a tertiary educational qualification (76.2%). Most participants reported New Zealand European ethnicity ($n = 80$, 79.2%).

Table 1 displays current and pre-earthquake mental disorders from the MINI. Most treatment-seeking participants had earthquake-related PTSD (79.0%). The remainder (21.0%) did not meet full criteria for PTSD and were diagnosed with adjustment disorder with earthquake-related anxiety. Most treatment-seeking participants had at least one pre-earthquake mental disorder (79.9% of the 174 participants with available pre-earthquake MINI data), with mood disorders (59.1%), generalised anxiety disorder (31.5%), and panic disorder (29.9%) the most common. Rates of current co-occurring disorders were high in this group, with mood disorders (74.0%), generalised anxiety disorder (48.6%), and panic disorder (45.9%) the most common. Low rates of current disorders were found in the control group. Six participants had a current disorder (four with agoraphobia, one with panic disorder, and one with a substance use disorder) that pre-dated the earthquakes. Thirty-seven percent had a pre-earthquake mood disorder and 33.0% had a pre-earthquake substance use disorder. Rates of other pre-earthquake disorders ranged from 1.1% for adjustment disorder and generalised anxiety disorder to 14.6% for PTSD.

Group Differences in Current Psychological Functioning

Table 2 displays summary statistics and results from univariate logistic regression analyses for variables related to current psychological functioning. Compared to control participants, treatment-seeking participants reported greater PTSD symptom severity (PCL-S), less resilience (CD-RISC), greater depression, anxiety, and stress symptoms (DASS), poorer social adjustment (SAS), and greater aggression (BPAQ). All current psychological functioning factors were significantly associated with group membership (all $p < .001$).

Table 1

Current and Pre-Earthquake Disorders from the MINI for the Treatment-Seeking and Control Groups

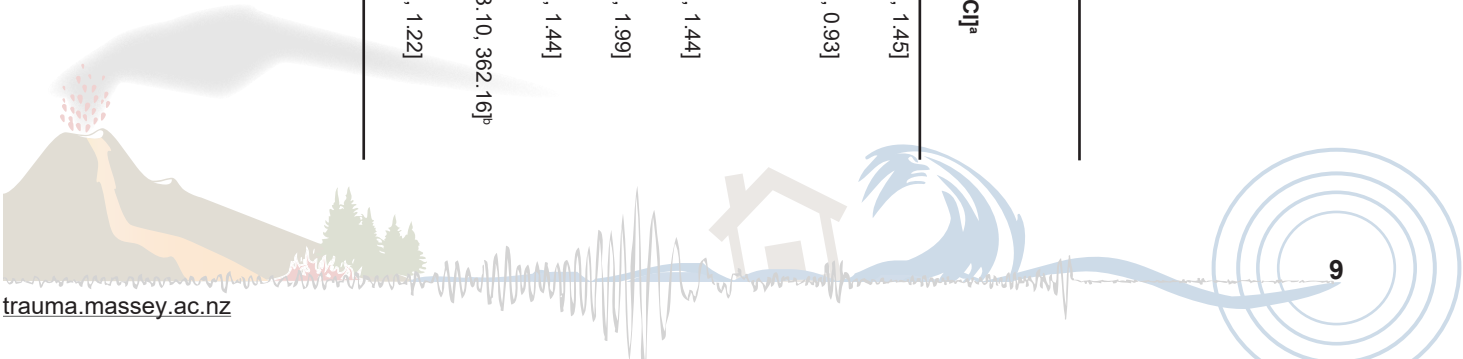
Disorder	Treatment-seeking group ^a		Control group ^b	
	<i>n</i>	%	<i>n</i>	%
PTSD				
Current	143	79.0	0	0.0
Pre-earthquake	46	26.1	13	14.6
Adjustment disorder				
Current	39	21.5	0	0.0
Pre-earthquake	4	2.3	1	1.1
Panic disorder				
Current	83	45.9	1	1.0
Pre-earthquake	53	29.9	5	5.7
Agoraphobia				
Current	48	26.5	4	4.0
Pre-earthquake	29	16.3	5	5.7
Social phobia				
Current	49	27.1	0	0.0
Pre-earthquake	46	25.8	5	5.7
Obsessive compulsive disorder				
Current	16	8.8	0	0.0
Pre-earthquake	16	9.0	3	3.4
Generalised anxiety disorder				
Current	88	48.6	0	0.0
Pre-earthquake	56	31.5	1	1.1
Mood disorders^c				
Current	134	74.0	0	0.0
Pre-earthquake	104	59.1	32	36.8
Substance use disorders				
Current	18	9.9	1	1.0
Pre-earthquake	24	13.4	29	33.0

Note. Low levels of current disorders in the control group were due to exclusion criteria for control participants. ^a *n* ranged from 176 to 181, ^b *n* ranged from 87 to 101, ^c Major depression or dysthymia

Table 2
Descriptive Statistics and Results of Univariate Logistic Regression Analyses Examining Group Differences in Current Psychological Functioning

	Treatment-seeking group		Control group		B	SE	p	OR [95% CI] ^a
	Mean	SD	Mean	SD				
Posttraumatic Stress Disorder Checklist (<i>n</i> = 219)	54.70	14.93	21.66	4.89	0.28	0.05	< .001	1.33 [1.21, 1.45]
Connor-Davidson Resilience Scale (<i>n</i> = 216)	52.81	18.09	75.83	12.03	-0.10	0.01	< .001	0.91 [0.88, 0.93]
Depression Anxiety Stress Scales								
Depression subscale score (<i>n</i> = 284)	21.34	11.92	2.67	3.69	0.29	0.04	< .001	1.33 [1.24, 1.44]
Anxiety subscale score (<i>n</i> = 284)	21.45	10.54	1.19	2.42	0.54	0.08	< .001	1.71 [1.47, 1.99]
Stress subscale score (<i>n</i> = 284)	26.33	9.66	5.15	5.09	0.29	0.04	< .001	1.34 [1.25, 1.44]
Social Adjustment Scale (<i>n</i> = 217)	2.43	0.51	1.63	0.29	4.70	0.61	< .001	109.48 [33.10, 362.16] ^b
Buss-Perry Aggression Questionnaire (<i>n</i> = 219)	23.44	9.43	15.58	5.51	0.15	0.03	< .001	1.16 [1.10, 1.22]

Note. ^a ORs relate to a 1-unit increase on each scale. ^b OR is large due to the small range associated with this variable



Main Analyses

Univariate Logistic Regression Analyses

Table 3 displays summary statistics and results from univariate logistic regression analyses comparing the groups on gender, age, education level, relationship status, number of pre-earthquake mental disorders (MINI), objective earthquake exposure severity and exposure-related distress (TESS), peritraumatic distress (PDI), number of life events experienced in the past 5 years, and difficulty associated with life events (LES). The treatment-seeking group was significantly younger than the control group ($p < .001$) and was characterised by a greater proportion of female participants ($p = .001$), and a smaller proportion of participants with a tertiary educational qualification ($p < .001$) and participants in a relationship ($p = .006$). Treatment-seeking and control participants reported comparable degrees of objective earthquake exposure ($p = .058$). Treatment-seeking participants had significantly more pre-earthquake disorders, reported greater exposure-related distress (TESS), greater peritraumatic distress (PDI), and greater exposure to and difficulty associated with life events (LES) compared to control participants (LES; all $p \leq .001$).

Multivariate Logistic Regression Analyses

Using a significance threshold of $p < .10$, the following factors were entered as independent variables into multivariate logistic regression analyses to identify factors uniquely associated with group membership: age, gender, education level, relationship status, number of pre-earthquake disorders, objective earthquake exposure severity and exposure-related distress, peritraumatic distress, number of life events in past 5 years, and difficulty associated with life events. Both forwards and backwards models converged to the same significant ($p < .05$) factors, education level (proportion of participants with a tertiary educational qualification) and peritraumatic distress (as measured by the PDI). The final model including education level and peritraumatic distress (with 116 treatment-seeking participants and 101 control participants) was statistically significant ($\chi^2(2) = 139.02$, $p < .001$). Having a tertiary educational qualification was associated with decreased odds of membership in the treatment-seeking group (OR = 0.26, 95% CI = 0.12, 0.60). Greater peritraumatic distress was associated with membership in the treatment-seeking group. For every 1-point increase in PDI score, the odds of treatment-seeking increased approximately 8.0 times (OR = 8.08, 95% CI = 4.68, 13.94). This final

model including education level and the PDI total score demonstrated excellent predictive accuracy (AUROC = .91, $p < .001$, 95% CI = .87, .95). Adjusting the final model for time elapsed between the onset of the earthquakes and assessment (less than 2.5 years vs. more than 2.5 years) did not substantially alter the results of the model.

Some treatment-seeking participants did not have available data for education level ($n = 32$) or the PDI ($n = 68$). Treatment-seeking participants with and without available education and PDI data were compared on several factors for which complete, or almost complete, data were available, including some sociodemographic (gender, age, ethnicity) and clinical (DASS depression, anxiety, and stress subscales, number of current disorders) characteristics. Treatment-seeking participants with and without available education and PDI data did not differ significantly on any of these variables (all $p \geq .259$), suggesting treatment-seeking participants with available education and PDI data were comparable to those without available data for these variables.

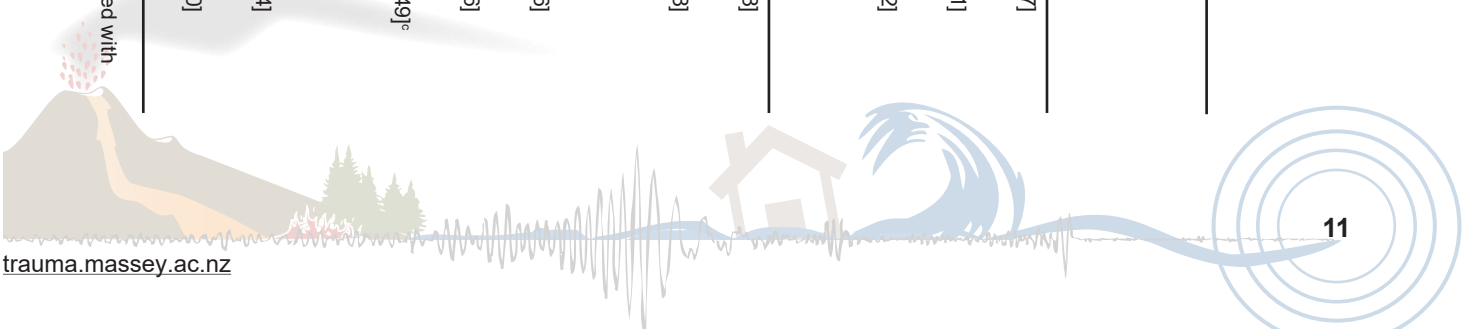
Discussion

The current study identified factors distinguishing individuals who experienced distinct outcomes following exposure to the Canterbury earthquakes: those who sought treatment for severe earthquake-related distress, and those who did not develop earthquake-related mental health issues and self-identified as coping well. Univariate analyses indicated the groups experienced comparable degrees of objective earthquake exposure, but treatment-seeking participants reported more pre-earthquake mental disorders, greater exposure-related distress and peritraumatic distress, more life events experienced in the past 5 years, and greater difficulty associated with life events. Treatment-seeking participants were also significantly younger on average, were more likely to be female, less likely to have a tertiary educational qualification, and less likely to be in a relationship. In multivariate analyses, only peritraumatic distress and education level were uniquely associated with group membership, suggesting the remaining factors were not associated with post-trauma outcome beyond their association with peritraumatic distress and education level. These two variables demonstrated excellent predictive accuracy, indicating peritraumatic distress and education level were important factors distinguishing treatment-seeking and control participants.

Table 3
Descriptive Statistics and Results of Main Univariate Logistic Regression Analyses

	Treatment-seeking group		Control group		B	SE	p	OR [95% CI] ^b
	n	%	n	%				
Gender (female; <i>n</i> = 284) ^a	153	83.6	67	66.3	0.95	0.29	.001	2.59 [1.47, 4.57]
Education (with tertiary qualification; <i>n</i> = 253)	55	36.2	77	76.2	-1.73	0.29	<.001	0.18 [0.10, 0.31]
Relationship status (in a relationship; <i>n</i> = 284)	96	52.5	70	69.3	-0.72	0.26	.006	0.49 [0.29, 0.82]
	Mean	SD	Mean	SD	B	SE	p	OR [95% CI] ^b
Age (years; <i>n</i> = 285)	43.74	13.27	50.13	11.02	-0.04	0.01	<.001	0.96 [0.94, 0.98]
Number of pre-earthquake disorders (<i>n</i> = 260)	2.21	1.92	1.28	1.50	0.34	0.09	<.001	1.40 [1.17, 1.68]
Traumatic Exposure Severity Scale								
Occurrence score (<i>n</i> = 214)	12.56	3.83	11.58	3.55	0.07	0.04	.058	1.07 [1.00, 1.16]
Distress score (<i>n</i> = 214)	49.54	22.78	33.79	17.34	0.04	0.01	<.001	1.04 [1.02, 1.06]
Peritraumatic Distress Inventory (<i>n</i> = 217)	2.48	0.86	1.04	0.63	2.19	0.28	<.001	8.95 [5.17, 15.49] ^c
Life Events Scale								
Number of life events in past 5 years (<i>n</i> = 215)	11.61	6.73	8.84	5.10	0.08	0.03	.001	1.08 [1.03, 1.14]
Difficulty score (<i>n</i> = 215)	2.65	1.11	2.08	0.67	0.82	0.19	<.001	2.26 [1.55, 3.30]

Note. ^a Analysis excludes one participant who identified as a transgender woman. ^b ORs relate to a 1-unit increase on each scale. ^c ORs is large due to the small range associated with this variable.



The finding that emotional and physiological distress experienced during the earthquakes (i.e., peritraumatic distress) emerged as an important distinguishing factor is consistent with theoretical models of PTSD highlighting the role of subjective experiences in the development of the disorder. Ehlers and Clark's (2000) cognitive model of PTSD posits that PTSD develops when a traumatic event is processed and interpreted in a way that perpetuates a sense of serious ongoing threat. Thus, the model emphasises subjective interpretations of the event and its sequelae as an important determinant of post-trauma outcomes. Existing theory suggests subjective peritraumatic responses contribute to the development of PTSD because they influence trauma-related appraisals and the nature of the trauma memory. People who experience strong peritraumatic responses may be more likely to interpret this as evidence for a negative view of themselves (e.g., "I cannot cope with stress"), creating an internal threat to their sense of self as capable and worthy. Additionally, strong peritraumatic responses may prompt data-driven processing (processing of sensory/perceptual characteristics of the event), rather than conceptual processing (processing of the event's meaning) during traumatic events, which may interfere with the encoding of the event and result in trauma memories that are poorly integrated with the wider autobiographical memory network. Poorly integrated trauma memories are thought to contribute to a sense of ongoing threat because they underlie the development of the re-experiencing symptoms (e.g., flashbacks) characteristic of PTSD (Ehlers & Clark, 2000). Biological models suggest peritraumatic distress contributes to increased risk for PTSD through fear conditioning. Strong peritraumatic reactions elicit biological fear responses, resulting in the formation of strong associations between trauma cues and fear responding (Pitman et al., 2012). Continued reactivation of the fear response through exposure to trauma-related cues may lead to sensitization of the neurocircuitry implicated in fear responding, which may contribute to increased risk for PTSD (Perusini et al., 2016).

Findings of the current study are also consistent with existing empirical research. Meta-analyses indicate strong peritraumatic responses are associated with the development of PTSD (Ozer et al., 2003; Vance et al., 2018). Importantly, peritraumatic distress has been found to be uniquely associated with PTSD symptoms in studies controlling for degree of trauma exposure (Cénat & Derivois, 2014; Shigemura et al., 2014), suggesting the association between subjective peritraumatic responses

and PTSD symptoms persists among people exposed to similar degrees of objective exposure. Prior studies of the Canterbury earthquakes have also found subjective responses to be an important determinant of post-trauma outcomes, consistent with the current findings. Boden et al. (2015) investigated the associations between degree of earthquake exposure, peritraumatic distress, distress caused by ongoing earthquake-related disruption, and PTSD symptoms among a birth cohort exposed to the Canterbury earthquakes. Using structural equation modelling, peritraumatic distress and disruption distress were found to fully mediate the association between degree of earthquake exposure and PTSD symptoms, suggesting subjective emotional experiences, both during the earthquakes and in response to ongoing disruption, are important factors contributing to the development of PTSD symptoms. Furthermore, these findings were robust to the inclusion of a range of covariates, including gender, ethnicity, prior mental disorders, and intellectual functioning. Peritraumatic distress was also found to be an important determinant of post-earthquake depression symptoms in a similar study of the same participants (Bell et al., 2017), suggesting peritraumatic distress influences mental health outcomes beyond PTSD symptoms.

Some evidence suggests recall of peritraumatic responses is influenced by current symptom levels; those with PTSD may attribute their symptoms to their experience of the traumatic event, resulting in overestimates of peritraumatic distress (Candel & Merckelbach, 2004). Thus, it is possible this finding reflects group membership, rather than the role of peritraumatic distress in the development of post-trauma outcomes. However, other evidence suggests memories of traumatic events are recalled reliably over time, even more so than memories for positive events, and that reliability is not influenced by the presence of PTSD (Porter & Peace, 2007).

Although a large body of research supports the association between peritraumatic responses and the development of PTSD, the evidence is not unequivocal. For example, van der Velden and Wittman (2008) conducted a systematic review of prospective studies examining the association between peritraumatic dissociation and the development of subsequent PTSD symptoms. Peritraumatic dissociation occurs when an individual feels detached from their surroundings or sense of self during a traumatic event, and has been linked to the development of poorer post-trauma outcomes (Lensvelt-Mulders et al., 2008). Although several studies reported statistically significant bivariate

associations, few studies found peritraumatic dissociation to be an independent predictor of subsequent PTSD symptoms when controlling for other factors, suggesting peritraumatic dissociation has little independent predictive value. Similarly, diagnostic criteria for PTSD no longer include subjective responses to qualify events as traumatic. In DSM-IV, Criterion A2 specified that the individual must have responded to the traumatic event with intense fear, helplessness, or horror (American Psychiatric Association, 2000). This criterion was removed in DSM-5, partly due to research suggesting the addition of subjective peritraumatic responses (Criterion A2) does not increase predictive utility for PTSD beyond endorsement of experiencing a traumatic event (Criterion A1; Bedard-Gilligan & Zoellner, 2008). Evidence suggests there are distinct peritraumatic processes that could be differentially associated with PTSD symptom clusters. Massazza et al. (2021) investigated the factor structure of items from six measures of peritraumatic responding and identified five underlying factors including mental defeat, somatoform dissociation, cognitive overload, immobility, and distress. Different factors were significant predictors for different symptoms clusters, which could partially account for the mixed findings regarding the role of peritraumatic responses in the development of PTSD.

Unlike peritraumatic distress, exposure-related distress (as assessed by the TESS) was significantly associated with group membership in univariate, but not multivariate, analyses in the current study. This finding suggests exposure-related distress was not independently associated with post-trauma outcome when accounting for peritraumatic distress and education level. The measure of exposure-related distress used in the current study captured distress in response to acute experiences (e.g., sustaining injury, being buried under rubble), but also distress associated with ongoing earthquake-related disruption (e.g., damage to home, loss of employment). Thus, this finding is somewhat inconsistent with prior studies identifying subjective responses to ongoing disruption, in addition to peritraumatic responses, as being uniquely associated with PTSD symptoms (e.g., Boden et al., 2015). Findings suggest that while treatment-seeking participants experienced greater levels of exposure-related distress compared to control participants (who coped well), subjective peritraumatic responses were relatively more important in determining post-trauma outcomes.

Education level also emerged as an important factor distinguishing treatment-seeking and control participants; treatment-seeking participants were less likely to

have a tertiary educational qualification compared to those who coped well. Meta-analyses have identified lower education level as a risk factor for PTSD among adults following earthquakes, consistent with the current findings (Tang et al., 2017). However, the high proportion of participants with a tertiary educational qualification in the control group (76%) may reflect the convenience sampling strategy used to recruit these participants. Lower education level may act as a proxy for lower socioeconomic status. Individuals of lower socioeconomic status may have fewer resources (e.g., financial resources) available to help cope with ongoing earthquake-related stressors (e.g., property damage), which may increase risk for the development of PTSD symptoms (Hobfoll et al., 2006).

Number of pre-earthquake mental disorders, number of life events experienced in the past 5 years, and difficulty associated with life events, as well as the remaining sociodemographic factors of interest (gender, age, and relationship status) were significantly associated with post-trauma outcome in univariate analyses but did not uniquely distinguish participants in multivariate models. This is inconsistent with prior studies in which factors relating to pre-trauma mental disorders (Ogle et al., 2016), life stress (Bonanno et al., 2007), and sociodemographic characteristics (particularly female gender; Wang et al., 2011) have been identified as unique predictors of PTSD symptoms. However, it is difficult to make direct comparisons between these findings and the current findings, as prior studies have controlled for different sets of variables in multivariate analyses. Of note, studies reporting unique associations between pre-trauma mental disorder and life stress and PTSD symptoms have not controlled for peritraumatic distress (e.g., Bonanno et al., 2007; Ogle et al., 2016), which could partially account for discrepancies between the current findings and those of prior studies. Peritraumatic distress may be an important factor to control for statistically in future studies examining factors associated with post-trauma outcomes.

Results of the current study have important potential clinical applications. Peritraumatic distress and education level emerged as important factors distinguishing treatment-seeking and control participants with moderate to high degrees of earthquake exposure, suggesting these factors would be useful in screening for vulnerable individuals following potentially traumatic events. Screening for subjective peritraumatic responses could be particularly important in the context of disasters, which often result in large numbers of people with high

levels of objective exposure. Screening for subjective peritraumatic responses has potential to improve the accuracy of screening efforts and ensure that resources, which are often limited in post-disaster periods, are used efficiently.

The current study should be interpreted considering the following limitations. Treatment-seeking participants were drawn from a clinical service population and may not be representative of all Canterbury residents who sought treatment for PTSD following the earthquakes. Although the service was the main provider of publicly-funded mental health treatment for those with severe earthquake-related distress, some people received treatment through other services (e.g., private psychologists and psychiatrists). Furthermore, treatment-seeking participants may not be representative of all individuals who developed PTSD after the earthquakes, given some people are likely to have not sought treatment at all. For instance, some research indicates men are less likely to seek treatment following trauma exposure than women (Gavrilovic et al., 2005), suggesting women may have been overrepresented in the treatment-seeking sample. Additionally, a convenience sampling strategy was used to recruit control participants, so these participants may not have been demographically representative of all Canterbury residents who coped well following the earthquakes. For instance, the control group was characterised by a large proportion of people with a tertiary educational qualification, which may have partially accounted for the emergence of education level as an important distinguishing factor. Additionally, New Zealand European participants were overrepresented in both the treatment-seeking and control groups. Thus, results here should be interpreted with caution. Future studies are needed to determine the generalisability of these findings to other populations, and could clarify why some established predictors of post-trauma outcome did not emerge as significant distinguishing factors in the current study. The current study was cross-sectional. Post-trauma trajectories are dynamic processes and cross-sectional studies may fail to capture important changes occurring over time. The use of retrospective self-reports may have compromised the accuracy of some self-report measures, such as peritraumatic distress, and pre-earthquake MINI disorders, which were based on post-earthquake recall. Some factors previously found to be associated with post-trauma outcomes (e.g., social support, prior trauma exposure) were not assessed in this study and thus were not included in multivariate models. Some participants did not have available data for all variables, which may have

reduced power to detect effects.

The current study also has considerable strengths. We compared participants who experienced markedly different outcomes following similar degrees of objective exposure to the same event, allowing for an examination of factors associated with post-trauma outcomes among people with similar objective experiences. Control participants were recruited into the study only if they had experienced substantial earthquake exposure, ensuring these participants were not coping well simply due to lower levels of exposure severity. Clinician-administered interviews were used to assess the presence of current and pre-earthquake mental disorders, ensuring this information was of high quality. Well-validated measures were used to assess participants' current psychological functioning.

To conclude, the current study identified factors distinguishing treatment-seeking individuals from those who did not develop any mental health issues and self-identified as having coped well following a major earthquake. Findings suggested that while the groups experienced comparable degrees of objective earthquake exposure, treatment-seeking participants reported higher levels of peritraumatic distress and lower levels of education, which potentially contributed to the development of distinct post-earthquake outcomes. The findings here suggest subjective peritraumatic responses and education level could be included in screening efforts following disasters.

Authors' Note. The study was approved by the National Health and Disability Ethics Committee (URA/11/EXP/027, URA/12/03/011). All participants gave written informed consent. Data are not publicly available due to privacy/ethical restrictions. This research received no specific grant from any funding agency, commercial or not-for-profit sectors.

References

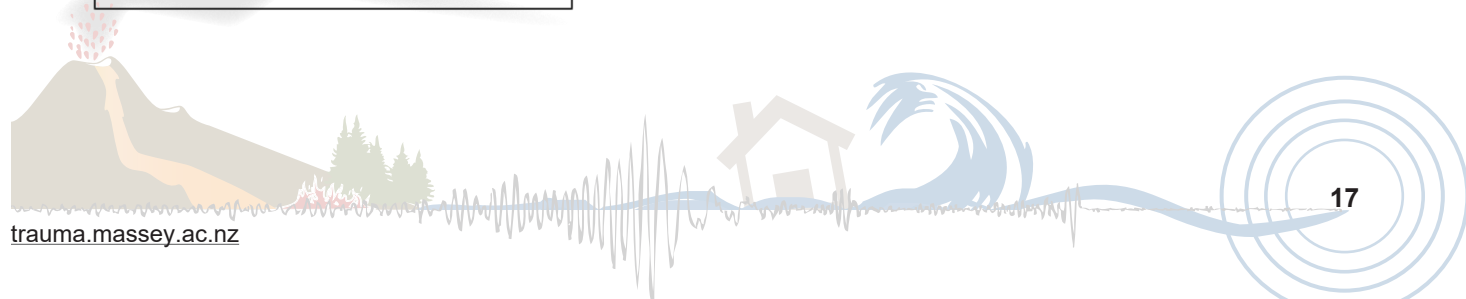
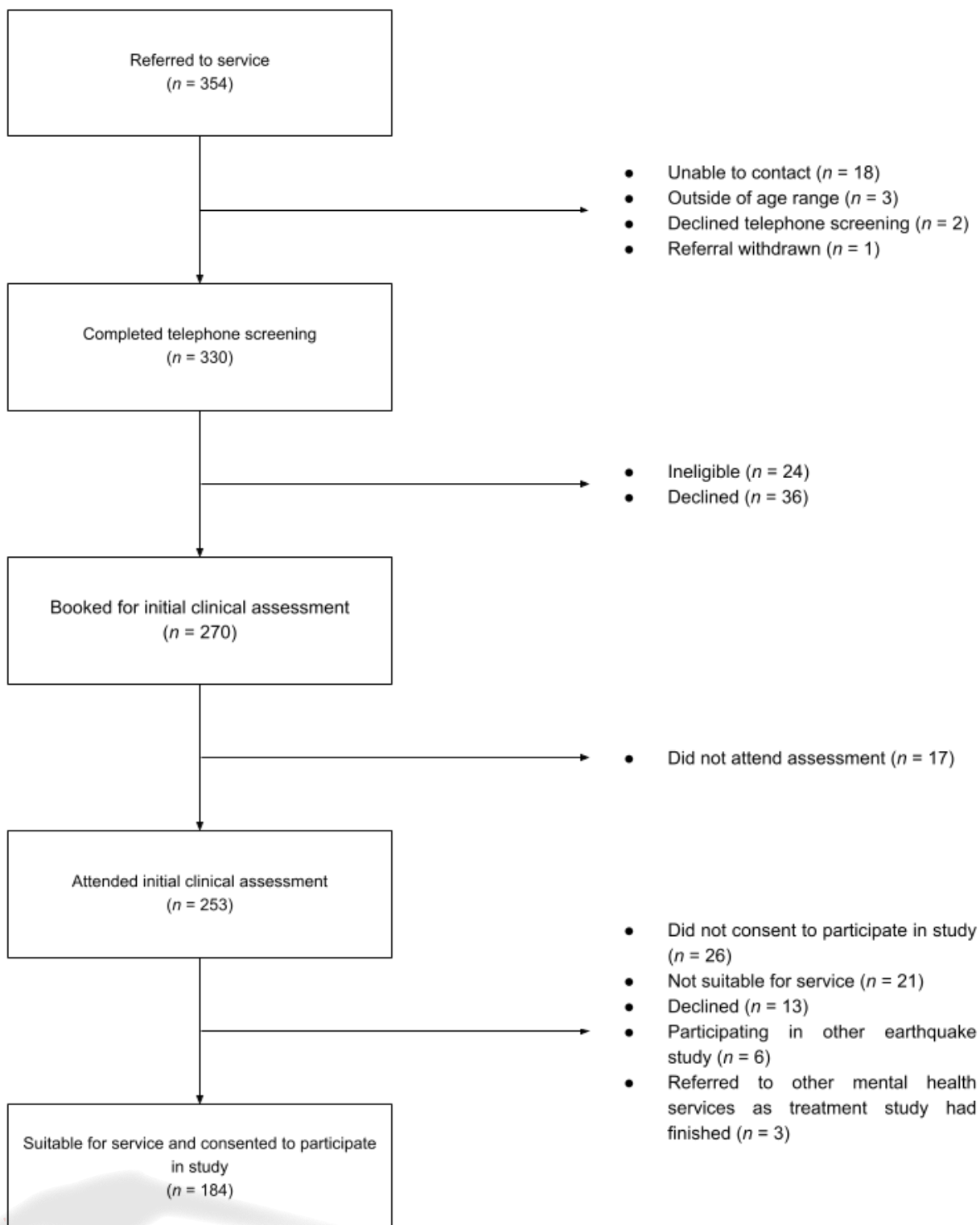
- American Psychiatric Association. (2000). *Diagnostic and statistical manual of mental disorders* (4th ed ed.). American Psychiatric Publishing, Inc.
- Ardagh, M. W., Richardson, S. K., Robinson, V., Than, M., Gee, P., Henderson, S., Khodaverdi, L., McKie, J., Robertson, G., Schroeder, P. P., & Deely, J. M. (2012). The initial health-system response to the earthquake in Christchurch, New Zealand, in February, 2011. *The Lancet*, 379(9831), 2109-2115. [https://doi.org/10.1016/S0140-6736\(12\)60313-4](https://doi.org/10.1016/S0140-6736(12)60313-4)
- Asmundson, G. J. G., Frombach, I., McQuaid, J., Pedrelli, P., Lenox, R., & Stein, M. B. (2000). Dimensionality of posttraumatic stress symptoms: A confirmatory factor analysis of DSM-IV symptom clusters and other symptom models. *Behaviour Research and Therapy*, 38(2), 203-214.

- [https://doi.org/10.1016/S0005-7967\(99\)00061-3](https://doi.org/10.1016/S0005-7967(99)00061-3)
- Asnaani, A., & Hall-Clark, B. (2017). Recent developments in understanding ethnocultural and race differences in trauma exposure and PTSD. *Current Opinion in Psychology*, 14, 96-101. <https://doi.org/10.1016/j.copsyc.2016.12.005>
- Auxéméry, Y. (2018). Post-traumatic psychiatric disorders: PTSD is not the only diagnosis. *La Presse Médicale*, 47(5), 423-430. <https://doi.org/10.1016/j.lpm.2017.12.006>
- Bedard-Gilligan, M., & Zoellner, L. A. (2008). The utility of the A1 and A2 criteria in the diagnosis of PTSD. *Behaviour Research and Therapy*, 46(9), 1062-1069. <https://doi.org/10.1016/j.brat.2008.06.009>
- Bell, C. J., Boden, J. M., Horwood, L. J., & Mulder, R. T. (2017). The role of peri-traumatic stress and disruption distress in predicting symptoms of major depression following exposure to a natural disaster. *Australian and New Zealand Journal of Psychiatry*, 51(7), 711-718. <https://doi.org/10.1177/0004867417691852>
- Benjet, C., Bromet, E., Karam, E. G., Kessler, R. C., McLaughlin, K. A., Ruscio, A. M., Shahly, V., Stein, D. J., Petukhova, M., Hill, E., Alonso, J., Atwoli, L., Bunting, B., Bruffaerts, R., Caldas-de-Almeida, J. M., de Girolamo, G., Florescu, S., Gureje, O., Huang, Y., . . . Koenen, K. C. (2015). The epidemiology of traumatic event exposure worldwide: Results from the World Mental Health Survey Consortium. *Psychological Medicine*, 46(2), 327-343. <https://doi.org/10.1017/S0033291715001981>
- Blanchard, E. B., Jones-Alexander, J., Buckley, T. C., & Forneris, C. A. (1996). Psychometric properties of the PTSD Checklist (PCL). *Behaviour Research and Therapy*, 34(8), 669-673. [https://doi.org/10.1016/0005-7967\(96\)00033-2](https://doi.org/10.1016/0005-7967(96)00033-2)
- Boden, J. M., Fergusson, D. M., Horwood, L. J., & Mulder, R. T. (2015). The role of peri-traumatic stress and disruption distress in predicting post-traumatic stress disorder symptoms following exposure to a natural disaster. *British Journal of Psychiatry Open*, 1(1), 81-86. <https://doi.org/10.1192/bjpo.bp.115.001180>
- Bonanno, G. A. (2004). Loss, trauma, and human resilience: Have we underestimated the human capacity to thrive after extremely aversive events? *The American Psychologist*, 59(1), 20-28. <https://doi.org/10.1037/0003-066X.59.1.20>
- Bonanno, G. A., Galea, S., Bucciarelli, A., & Vlahov, D. (2006). Psychological resilience after disaster: New York city in the aftermath of the September 11th terrorist attack. *Psychological Science*, 17(3), 181-186. <https://doi.org/10.1111/j.1467-9280.2006.01682.x>
- Bonanno, G. A., Galea, S., Bucciarelli, A., & Vlahov, D. (2007). What predicts psychological resilience after disaster? The role of demographics, resources, and life stress. *Journal of Consulting and Clinical Psychology*, 75(5), 671-682. <https://doi.org/10.1037/0022-006X.75.5.671>
- Brewin, C. R., Andrews, B., & Valentine, J. D. (2000). Meta-analysis of risk factors for posttraumatic stress disorder in trauma-exposed adults. *Journal of Consulting and Clinical Psychology*, 68(5), 748-766. <https://doi.org/10.1037/0022-006X.68.5.748>
- Brunet, A., Weiss, D. S., Metzler, T. J., Best, S. R., Neylan, T. C., Rogers, C., Fagan, J., & Marmar, C. R. (2001). The Peritraumatic Distress Inventory: A proposed measure of PTSD criterion A2. *American Journal of Psychiatry*, 158(9), 1480-1485. <https://doi.org/10.1176/appi.ajp.158.9.1480>
- Bryant, F. B., & Smith, B. D. (2001). Refining the architecture of aggression: A measurement model for the Buss-Perry Aggression Questionnaire. *Journal of Research in Personality*, 35(2), 138-167. <https://doi.org/10.1006/jrpe.2000.2302>
- Buss, A. H., & Perry, M. (1992). The Aggression Questionnaire. *Journal of Personality and Social Psychology*, 63(3), 452-459. <https://doi.org/10.1037/0022-3514.63.3.452>
- Candel, I., & Merckelbach, H. (2004). Peritraumatic dissociation as a predictor of post-traumatic stress disorder: A critical review. *Comprehensive Psychiatry*, 45(1), 44-50. <https://doi.org/10.1016/j.comppsy.2003.09.012>
- Cénat, J. M., & Derivois, D. (2014). Assessment of prevalence and determinants of posttraumatic stress disorder and depression symptoms in adults survivors of earthquake in Haiti after 30 months. *Journal of Affective Disorders*, 159, 111-117. <https://doi.org/10.1016/j.jad.2014.02.025>
- Christiansen, D. M., & Hansen, M. (2015). Accounting for sex differences in PTSD: A multi-variable mediation model. *European Journal of Psychotraumatology*, 6, Article 26068. <https://doi.org/10.3402/ejpt.v6.26068>
- Connor, K. M., & Davidson, J. R. T. (2003). Development of a new resilience scale: The Connor-Davidson Resilience Scale (CD-RISC). *Depression and Anxiety*, 18(2), 76-82. <https://doi.org/10.1002/da.10113>
- Ehlers, A., & Clark, D. M. (2000). A cognitive model of posttraumatic stress disorder. *Behaviour Research and Therapy*, 38(4), 319-345. [https://doi.org/https://doi.org/10.1016/S0005-7967\(99\)00123-0](https://doi.org/https://doi.org/10.1016/S0005-7967(99)00123-0)
- Elal, G., & Slade, P. (2005). Traumatic Exposure Severity Scale (TESS): A measure of exposure to major disasters. *Journal of Traumatic Stress*, 18(3), 213-220. <https://doi.org/10.1002/jts.20030>
- Fernandez, C. A., Vicente, B., Marshall, B. D., Koenen, K. C., Arheart, K. L., Kohn, R., Saldivia, S., & Buka, S. L. (2016). Longitudinal course of disaster-related PTSD among a prospective sample of adult Chilean natural disaster survivors. *International Journal of Epidemiology*, 46(2), 440-452. <https://doi.org/10.1093/ije/dyw094>
- Gavrilovic, J. J., Schützwohl, M., Fazel, M., & Priebe, S. (2005). Who seeks treatment after a traumatic event and who does not? A review of findings on mental health service utilization. *Journal of Traumatic Stress*, 18(6), 595-605. <https://doi.org/10.1002/jts.20068>
- Hobfoll, S. E., Tracy, M., & Galea, S. (2006). The impact of resource loss and traumatic growth on probable PTSD and depression following terrorist attacks. *Journal of Traumatic Stress*, 19(6), 867-878. <https://doi.org/10.1002/jts.20166>
- Kilpatrick, D. G., Resnick, H. S., Milanak, M. E., Miller, M. W., Keyes, K. M., & Friedman, M. J. (2013). National estimates of exposure to traumatic events and PTSD prevalence using DSM-IV and DSM-5 criteria. *Journal of Traumatic Stress*, 26(5), 537-547. <https://doi.org/10.1002/jts.21848>
- Lensvelt-Mulders, G., van der Hart, O., van Ochten, J. M., van Son, M. J. M., Steele, K., & Breeman, L. (2008). Relations among peritraumatic dissociation and posttraumatic stress: A meta-analysis. *Clinical Psychology Review*, 28(7), 1138-1151. <https://doi.org/10.1016/j.cpr.2008.03.006>
- Lovibond, P. F., & Lovibond, S. H. (1995). The structure of negative emotional states: Comparison of the Depression Anxiety Stress Scales (DASS) with the Beck Depression and Anxiety Inventories. *Behaviour Research and Therapy*, 33(3), 335-343. [https://doi.org/10.1016/0005-7967\(94\)00075-U](https://doi.org/10.1016/0005-7967(94)00075-U)
- Massazza, A., Joffe, H., Hyland, P., & Brewin, C. R. (2021). The structure of peritraumatic reactions and their relationship with PTSD among disaster survivors. *Journal of Abnormal*

- Psychology*, 130(3), 248-259. <https://doi.org/10.1037/abn0000663>
- North, C. S., & Pfefferbaum, B. (2013). Mental health response to community disasters: A systematic review. *JAMA*, 310(5), 507-518. <https://doi.org/10.1001/jama.2013.107799>
- Ogle, C. M., Rubin, D. C., & Siegler, I. C. (2016). Accounting for posttraumatic stress disorder symptom severity with pre- and posttrauma measures: A longitudinal study of older adults. *Clinical Psychological Science*, 4(2), 272-286. <https://doi.org/10.1177/2167702615583227>
- Ozer, E. J., Best, S. R., Lipsey, T. L., & Weiss, D. S. (2003). Predictors of posttraumatic stress disorder and symptoms in adults: A meta-analysis. *Psychological Bulletin*, 129(1), 52-73. <https://doi.org/10.1037/0033-2909.129.1.52>
- Perusini, J. N., Meyer, E. M., Long, V. A., Rau, V., Nocera, N., Avershal, J., Maksymetz, J., Spigelman, I., & Fanselow, M. S. (2016). Induction and expression of fear sensitization caused by acute traumatic stress. *Neuropsychopharmacology*, 41(1), 45-57. <https://doi.org/10.1038/npp.2015.224>
- Pitman, R. K., Rasmusson, A. M., Koenen, K. C., Shin, L. M., Orr, S. P., Gilbertson, M. W., Milad, M. R., & Liberzon, I. (2012). Biological studies of post-traumatic stress disorder. *Nature Reviews Neuroscience*, 13(11), 769-787. <https://doi.org/10.1038/nrn3339>
- Porter, S., & Peace, K. A. (2007). The scars of memory. *Psychological Science*, 18(5), 435-441. <https://doi.org/10.1111/j.1467-9280.2007.01918.x>
- Shalowitz, M. U., Berry, C. A., Rasinski, K. A., & Dannhausen-Brun, C. A. (1998). A new measure of contemporary life stress: Development, validation, and reliability of the CRISYS. *Health Services Research*, 33(5), 1381-1402. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1070321/>
- Sheehan, D. V., Lecrubier, Y., Sheehan, K. H., Amorim, P., Janavs, J., Weiller, E., Hergueta, T., Baker, R., & Dunbar, G. C. (1998). The Mini-International Neuropsychiatric Interview (M.I.N.I.): The development and validation of a structured diagnostic psychiatric interview for DSM-IV and ICD-10. *The Journal of Clinical Psychiatry*, 59, 22-33.
- Shigemura, J., Tanigawa, T., Nishi, D., Matsuoka, Y., Nomura, S., & Yoshino, A. (2014). Associations between disaster exposures, peritraumatic distress, and posttraumatic stress responses in Fukushima nuclear plant workers following the 2011 nuclear accident: The Fukushima NEWS Project Study. *PLOS ONE*, 9(2), e87516. <https://doi.org/10.1371/journal.pone.0087516>
- Tang, B., Deng, Q., Glik, D., Dong, J., & Zhang, L. (2017). A meta-analysis of risk factors for post-traumatic stress disorder (PTSD) in adults and children after earthquakes. *International Journal of Environmental Research and Public Health*, 14(12), 1537. <https://doi.org/10.3390/ijerph14121537>
- Tortella-Feliu, M., Fullana, M. A., Pérez-Vigil, A., Torres, X., Chamorro, J., Littarelli, S. A., Solanes, A., Ramella-Cravaro, V., Vilar, A., González-Parra, J. A., Andero, R., Reichenberg, A., Mataix-Cols, D., Vieta, E., Fusar-Poli, P., Ioannidis, J. P. A., Stein, M. B., Radua, J., & Fernández de la Cruz, L. (2019). Risk factors for posttraumatic stress disorder: An umbrella review of systematic reviews and meta-analyses. *Neuroscience & Biobehavioral Reviews*, 107, 154-165. <https://doi.org/10.1016/j.neubiorev.2019.09.013>
- Van der Velden, P. G., & Wittmann, L. (2008). The independent predictive value of peritraumatic dissociation for PTSD symptomatology after type I trauma: A systematic review of prospective studies. *Clinical Psychology Review*, 28(6), 1009-1020. <https://doi.org/10.1016/j.cpr.2008.02.006>
- Vance, M. C., Kovachy, B., Dong, M., & Bui, E. (2018). Peritraumatic distress: A review and synthesis of 15 years of research. *Journal of Clinical Psychology*, 74(9), 1457-1484. <https://doi.org/10.1002/jclp.22612>
- Wang, B., Ni, C., Chen, J., Liu, X., Wang, A., Shao, Z., Xiao, D., Cheng, H., Jiang, J., & Yan, Y. (2011). Posttraumatic stress disorder 1 month after 2008 earthquake in China: Wenchuan earthquake survey. *Psychiatry Research*, 187(3), 392-396. <https://doi.org/10.1016/j.psychres.2009.07.001>
- Weissman, M. M., & Bothwell, S. (1976). Assessment of social adjustment by patient self-report. *Archives of General Psychiatry*, 33(9), 1111-1115. <https://doi.org/10.1001/archpsyc.1976.01770090101010>

Appendix A

Flow of Treatment-Seeking Participants Through the Recruitment Process



Appendix B

Flow of Control Participants Through the Recruitment Process

